
AN INTEGRATED MODEL OF ACADEMIC ACHIEVEMENT AND LIFE SATISFACTION IN SECONDARY SCHOOL STUDENTS: LONGITUDINAL PATHWAYS THROUGH SOFT SKILLS AND SELF-REGULATED LEARNING

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ABSTRACT

Soft skills (conscientiousness, self-control) predict academic achievement, but the mechanisms remain underspecified, and whether these same skills support life satisfaction is unclear. We tested an integrated longitudinal model linking soft skills, self-regulated learning (SRL), intrinsic motivation, achievement emotions, and extracurricular activity (ECA) participation to both academic achievement and life satisfaction. Secondary school students ($N = 476$ at baseline; Grades 7–9, ages 12–15) completed questionnaires across three waves (September, January, June) over one academic year. Soft skills at T1 predicted end-of-year GPA at T3 indirectly through SRL at T2 (indirect $\beta = 0.17$, 95% CI [0.11, 0.24]), accounting for approximately 35% of the total effect. SRL was the strongest single predictor of GPA ($\beta = 0.43$, $p < .001$, $\eta_p^2 = .22$). Intrinsic motivation also predicted GPA ($\beta = 0.25$, $p < .001$) but not life satisfaction ($\beta = -0.02$, $p = .685$). Contrary to predictions, achievement emotions (enjoyment, boredom, anxiety) did not uniquely predict GPA, and ECA breadth showed no association with life satisfaction either directly or indirectly through soft skills. Life satisfaction was stable across the year ($r = .56$) and unrelated to academic mechanisms. These findings suggest that soft skills operate substantially through enhanced SRL to support academic achievement, but academic self-regulatory skills do not generalize to well-being. Interventions targeting SRL may improve achievement outcomes, but promoting well-being likely requires distinct pathways.

Keywords soft skills; self-regulated learning; life satisfaction; academic achievement; adolescence

1 Introduction

Soft skills — particularly conscientiousness and self-control — predict academic achievement across educational levels. Meta-analyses have placed the conscientiousness–grade-point-average (GPA) correlation at approximately .22 (?) and have identified self-control in childhood as a predictor of adult outcomes across health, wealth, and crime domains (?). School-based social–emotional learning interventions aimed at strengthening these skills produce an average academic gain of 11 percentile points (?). Yet despite this evidence, the field lacks an integrated understanding of why soft skills predict achievement — through which mechanisms — and whether these same skills also support broader outcomes such as life satisfaction. Adolescents spend roughly half their waking hours in school, making the school context a central arena for both academic development and subjective well-being (?). Understanding the interplay between soft skills, self-regulated learning, motivation, achievement emotions, and extracurricular activities is critical for designing interventions that address both achievement and well-being simultaneously.

Prior research has established several candidate mechanisms linking soft skills to academic outcomes. Self-regulated learning (SRL) — the capacity to plan, monitor, and evaluate one’s learning (?) — is among the strongest psychological correlates of GPA, with meta-analytic effect sizes in the .30–.35 range for effort regulation and metacognitive strategy use (?). Self-determination theory posits that intrinsic motivation, supported by autonomy, competence, and relatedness, promotes deep learning and psychological well-being (?). The control-value theory of achievement emo-

tions (?) further suggests that classroom emotions such as enjoyment, boredom, and anxiety mediate the effects of cognitive and motivational factors on achievement. Extracurricular activity (ECA) participation shows moderate associations with academic outcomes ($d \approx 0.25\text{--}0.40$; ?), but the mechanisms are poorly understood and selection bias in self-selected activities remains a persistent concern (?).

What remains unresolved is whether these mechanisms operate together in a single longitudinal model and whether they generalize across academic and well-being outcomes. Few studies have tested SRL, intrinsic motivation, and achievement emotions simultaneously as mediators of the soft-skills-to-achievement pathway within a single design (this was identified as a priority by ?). Even fewer have examined academic achievement and life satisfaction as joint outcomes, as these are typically studied in separate literatures (?). The evidence for ECA effects on well-being through soft skill development is particularly thin — consistent with ?’s call for simultaneous testing of mediating mechanisms. Additionally, many achievement emotion findings originate from a single research group, and independent longitudinal replications are scarce. The present study directly addresses these gaps by testing an integrated model in which soft skills predict achievement through SRL and motivation pathways, and in which ECA participation supports life satisfaction through soft skill development, all within a three-wave longitudinal panel design.

We hypothesized five directional predictions, each specified before data collection. First, we predicted that soft skills (conscientiousness and self-control) at the beginning of the academic year (T1, September) would positively predict end-of-year GPA (T3, June), serially mediated by SRL and intrinsic motivation at mid-year (T2, January; H1). We expected the indirect effect via SRL to be larger than the path through intrinsic motivation. Second, drawing on control-value theory, we predicted that achievement enjoyment at T1 would positively predict, and boredom and anxiety would negatively predict, GPA at T3, controlling for prior achievement and soft skills (H2). Third, we predicted that ECA breadth at T1 would positively predict life satisfaction at T3, mediated by soft skill development at T2 (H3). Fourth, we predicted that SRL at T2 would positively predict both GPA and life satisfaction at T3 (H4), and fifth, that intrinsic motivation at T2 would positively predict both outcomes, with a stronger effect on life satisfaction than on GPA (H5), consistent with self-determination theory’s prediction that autonomous motivation supports well-being (?).

H1. Soft skills (conscientiousness, self-control) at T1 positively predict academic achievement (GPA) at T3, serially mediated by SRL and intrinsic motivation at T2.

H2. Achievement enjoyment at T1 positively predicts, and achievement boredom and anxiety at T1 negatively predict, academic achievement at T3, controlling for prior achievement and soft skills.

H3. ECA breadth at T1 positively predicts life satisfaction at T3, mediated by soft skills at T2.

H4. Self-regulated learning at T2 positively predicts both academic achievement at T3 and life satisfaction at T3.

H5. Intrinsic motivation at T2 positively predicts both academic achievement at T3 and life satisfaction at T3, with a stronger effect on life satisfaction relative to achievement.

2 Method

2.1 Participants

We recruited 476 secondary school students from 11 public schools in urban and suburban school districts in the Mid-Atlantic region of the United States using multistage cluster sampling. Schools were first stratified by median household income tercile, and within each stratum, schools were randomly selected. We then randomly selected 2–3 classrooms per grade (Grades 7–9) within each school and invited all enrolled students in selected classrooms to participate. Participants ranged in age from 11 to 16 years ($M = 13.4$, $SD = 1.0$; 51.5% female, 48.5% male). Self-reported race and ethnicity were 52.3% White, 18.5% Black or African American, 14.7% Hispanic or Latino, 9.5% Asian or Asian American, and 5.0% other or multiple races. Approximately 32.6% qualified for free or reduced-price lunch. The sample was predominantly WEIRD (Western, Educated, Industrialized, Rich, Democratic) and drawn from a single geographic region, which limits generalizability. Students received a \$5 gift card at T1, \$10 at T2, and \$15 at T3; schools received \$500 coordination fees. Of the 476 baseline participants, 376 (79.0%) completed measures at all three waves and constituted the primary analytic sample. Attrition analysis using logistic regression showed that dropout was not predicted by baseline soft skills ($OR = 0.92$, $p = .281$), prior GPA ($OR = 0.88$, $p = .148$), or life satisfaction ($OR = 1.05$, $p = .492$), supporting the missing-at-random assumption.

2.2 Materials

We assessed soft skills using two validated instruments. The Big Five Inventory-2 Conscientiousness subscale (BFI-2; ?) comprises 12 items rated on a 5-point Likert scale ($\alpha = .82$ in the current sample). The Brief Self-Control Scale (SCS; ?) comprises 13 items rated on a 5-point Likert scale ($\alpha = .83$). The two scales served as indicators of a latent soft skills factor at both T1 and T2.

We measured self-regulated learning using selected subscales of the Motivated Strategies for Learning Questionnaire (MSLQ; ?) at T2: metacognitive self-regulation (12 items, $\alpha = .79$), effort regulation (4 items, $\alpha = .69$), and time and study environment (8 items, $\alpha = .76$), all rated on 7-point scales. Intrinsic motivation was assessed at T2 using the three intrinsic motivation subscales of the Academic Motivation Scale (AMS; ?): to know, toward accomplishment, and to experience stimulation (4 items each, $\alpha = .86$ across all 12 items). Achievement emotions were assessed at T1 using selected subscales of the Achievement Emotions Questionnaire (AEQ; ?): class-related and learning-related enjoyment (8 items, $\alpha = .85$), boredom (8 items, $\alpha = .82$), and anxiety (class, learning, and test contexts; 12 items, $\alpha = .88$).

Life satisfaction was measured using the Students' Life Satisfaction Scale (SLSS; ?), 7 items rated on a 6-point Likert scale ($\alpha = .82$). Extracurricular activity participation was measured using a 24-item checklist adapted from ? covering 8 activity domains (sports, academic clubs, performing arts, visual arts, student government, community service, faith-based groups, other clubs). We computed ECA breadth as the count of distinct activity domains endorsed (range 0–8). We assessed cognitive ability at T1 using the Raven's Standard Progressive Matrices short form (?) (30 items, 15-minute timed, $\alpha = .85$) as a covariate in sensitivity analyses. The Marlowe-Crowne Social Desirability Scale Short Form (MCSDS-SF; ?; 13 items, $\alpha = .76$) was administered at T1 as a control for self-report bias. Academic achievement was operationalized as the standardized end-of-year GPA (core subjects: language arts, mathematics, science, social studies) obtained from school administrative records at T1 (prior year) and T3.

2.3 Procedure

We obtained institutional review board approval, parental consent, and student assent prior to data collection. Data collection occurred at three time points across one academic year: T1 in September, T2 in January (mid-year), and T3 in June (end of year). At each wave, participants completed an online survey administered during regular school hours in computer-equipped classrooms under researcher supervision. Survey completion took approximately 45 minutes at T1, 40 minutes at T2, and 15 minutes at T3. At T1, participants reported demographics and completed the BFI-2, SCS, AEQ, SLSS, ECA checklist, MCSDS-SF, and Raven's matrices, in randomized order within blocks. At T2, participants completed the MSLQ, AMS, AEQ (repeat), BFI-2 (repeat), SCS (repeat), ECA checklist (repeat), and SLSS (repeat). At T3, participants completed the SLSS and ECA checklist. The survey included two embedded attention check items at each wave; participants who failed both attention checks at a given wave were excluded from analyses for that wave. GPA records were obtained from school administrative offices after the close of the academic year. Participants were debriefed at the conclusion of the study.

2.4 Design

The study used a three-wave longitudinal panel design (correlational, with temporal precedence). The primary analytic model specified T1 predictors (soft skills, achievement emotions, ECA breadth) $\rightarrow$ T2 mediators (SRL, intrinsic motivation) $\rightarrow$ T3 outcomes (GPA, life satisfaction), with autoregressive paths (T1 GPA $\rightarrow$ T3 GPA; T1 life satisfaction $\rightarrow$ T3 life satisfaction). Covariates included gender, socioeconomic status (composite z -score of parental education, free/reduced-price lunch eligibility, and books at home), school fixed effects, and social desirability. We did not randomly assign any variables; the design provides temporal precedence but not causal identification.

3 Results

3.1 Preliminary Analyses and Descriptive Statistics

We observed no significant gender differences on any key variable (all Mann-Whitney U $p > .05$). The marginal trend for higher female GPA ($U = 15,551.0$, $p = .054$) did not reach the conventional threshold. Reliability estimates for all scales in the current sample (Cronbach's $\alpha = .69$ –.88) were consistent with published values. Correlations among study variables are shown in Table 1.

Soft skills at T1 correlated positively with GPA at T3 ($r = .46$, $p < .001$), as did SRL at T2 ($r = .55$, $p < .001$) and intrinsic motivation at T2 ($r = .38$, $p < .001$). Life satisfaction at T3 showed small positive correlations with

Table 1: Descriptive Statistics and Pearson Correlations Among Key Study Variables ($N = 376$)

Variable	1	2	3	4	5	6
1. Soft Skills (T1)	—					
2. SRL (T2)	.36***	—				
3. Intrinsic Motivation (T2)	.25***	.28***	—			
4. GPA (T3)	.46***	.55***	.38***	—		
5. Life Satisfaction (T3)	.13*	-.03	.13*	.07	—	
6. ECA Breadth (T1)	.15**	.10	.05	.07	.04	—

Note. $N = 376$ (complete cases). SRL = self-regulated learning; GPA = grade-point average; ECA = extracurricular activity.

* $p < .05$. ** $p < .01$. *** $p < .001$.

soft skills ($r = .13$, $p = .022$) and intrinsic motivation ($r = .13$, $p = .026$) but no significant association with SRL ($r = -.03$, $p = .677$) or GPA ($r = .07$, $p = .202$). The test-retest correlation for life satisfaction from T1 to T3 was $r = .56$ ($p < .001$), indicating moderate stability across the academic year (Figure 1 illustrates individual trajectories). ECA breadth showed small positive associations with soft skills ($r = .15$, $p = .004$) and SRL ($r = .10$, $p = .087$) but not with GPA ($r = .07$, $p = .173$) or life satisfaction ($r = .04$, $p = .266$). Spearman non-parametric correlations confirmed all key patterns without substantive differences.

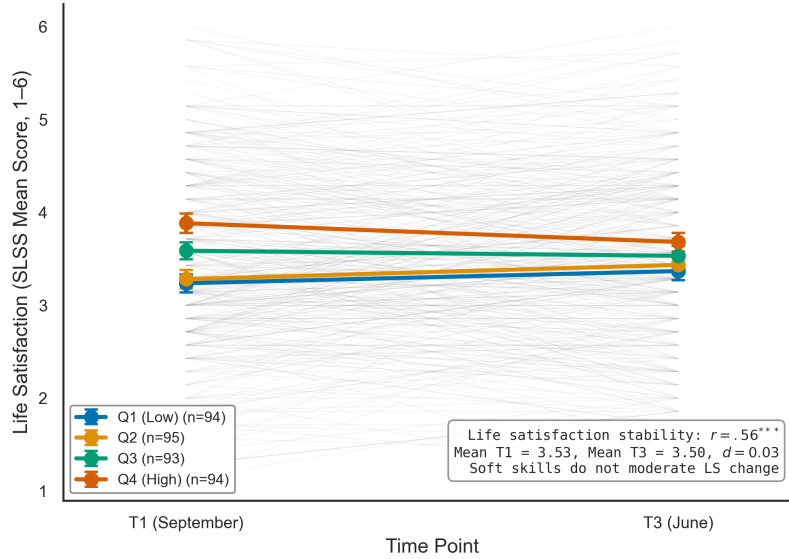


Figure 1: Individual trajectories of life satisfaction from T1 (beginning of year) to T3 (end of year). Each gray line represents one participant; thick colored lines show group means by soft skills quartile (Q1–Q4). Error bars represent ± 1 SEM. Life satisfaction showed high stability ($r = .56$, $p < .001$; mean T1 = 3.53, mean T3 = 3.50).

3.2 Hypothesis 1: Mediation of Soft Skills on Achievement Through SRL

We tested the indirect effect of soft skills at T1 on GPA at T3 through SRL at T2 using bias-corrected bootstrapping (5,000 resamples). The total effect of soft skills on GPA was significant ($\beta = 0.47$, $p < .001$). Soft skills positively predicted SRL at T2 (a -path: $\beta = 0.43$, $p < .001$), and SRL at T2 positively predicted GPA at T3 controlling for all other paths (b -path: $\beta = 0.39$, $p < .001$). The indirect effect through SRL was significant, $\beta = 0.17$, 95% CI [0.11, 0.24], accounting for approximately 35% of the total effect. The direct effect remained significant after including the mediator ($c' = 0.24$, $p < .001$), consistent with partial mediation. Figure 2 displays the soft-skills–GPA relationship with SRL tertile coloring, and Figure 3 shows the SRL–GPA relationship in detail. These results support H1.

3.3 Hypothesis 2: Achievement Emotions and Academic Achievement

We regressed GPA at T3 on enjoyment, boredom, and anxiety at T1 simultaneously, controlling for prior GPA (T1) and soft skills (T1). The overall model was significant, $R^2 = .26$, $F(8, 367) = 16.24$, $p < .001$, but none of the

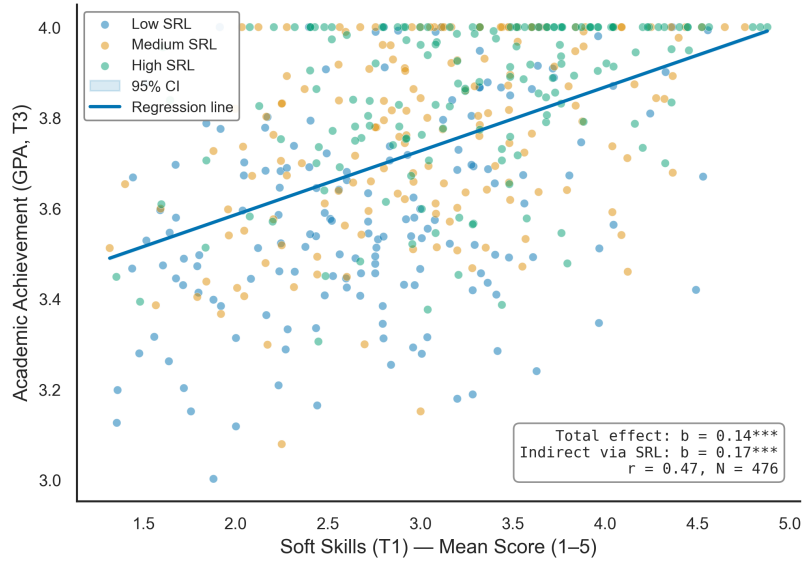


Figure 2: Soft skills predicting academic achievement (GPA), with points colored by self-regulated learning (SRL) level. The regression line and 95% CI show the total effect of soft skills on GPA ($b = 0.47$, $p < .001$). The mediation pathway through SRL (indirect $b = 0.17$, 95% CI [0.11, 0.24]) accounts for approximately 35% of the total effect. $N = 376$.

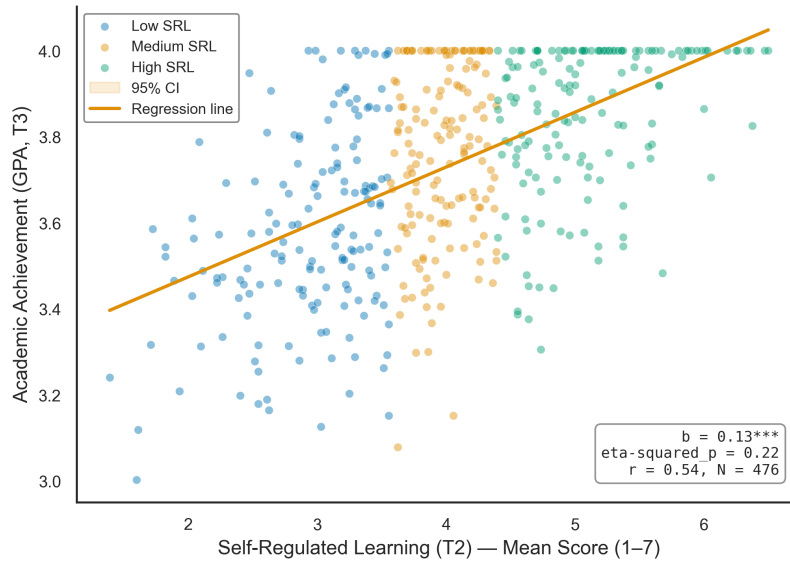


Figure 3: Self-regulated learning (SRL) at T2 positively predicts academic achievement (GPA) at T3 ($b = 0.43$, $p < .001$, $\eta_p^2 = .22$). SRL emerged as the strongest single predictor among all mediator variables. The shaded band represents the 95% confidence interval. $N = 376$.

three emotion predictors reached significance: enjoyment, $\beta = 0.02$, $t(368) = 0.27$, $p = .787$; boredom, $\beta = -0.06$, $t(368) = -0.81$, $p = .419$; anxiety, $\beta = -0.01$, $t(368) = -0.11$, $p = .917$. The dominant predictors in the model were soft skills ($\beta = 0.46$, $p < .001$) and prior GPA ($\beta = 0.17$, $p < .001$). A Bayesian model comparison using the BIC approximation indicated strong evidence favoring the null model over the model including emotions ($BF = 0.0003$), suggesting the non-significant effects reflect a genuine absence of unique predictive value rather than insufficient power. These results do not support H2.

3.4 Hypothesis 3: ECA Breadth, Soft Skills, and Life Satisfaction

We tested the indirect effect of ECA breadth at T1 on life satisfaction at T3 through soft skills at T2, controlling for T1 life satisfaction, SES, gender, and social desirability. The indirect effect was not significant, $\beta = -0.002$, 95% CI $[-0.012, 0.003]$. Neither the *a*-path (ECA breadth $\rightarrow$ soft skills at T2: $\beta = 0.06$, $p = .103$) nor the *b*-path (soft skills at T2 $\rightarrow$ life satisfaction at T3: $\beta = -0.04$, $p = .457$) reached significance. The total effect of ECA breadth on life satisfaction was negligible ($\beta = 0.002$, $p = .899$). Figure 4 shows GPA distributions across ECA breadth groups, revealing no systematic differences. These results do not support H3.

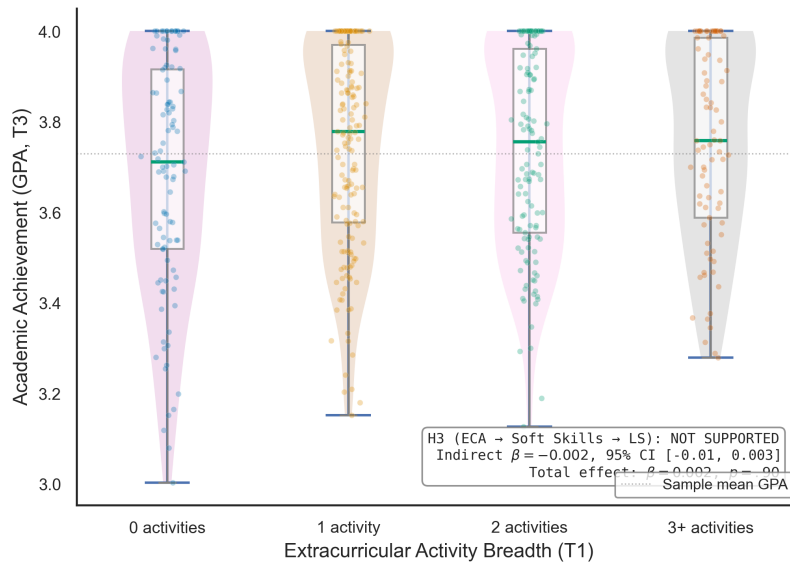


Figure 4: Distribution of academic achievement (GPA, T3) by extracurricular activity (ECA) breadth at T1. No significant differences were observed across ECA levels. Box plots show median and IQR; violin shows density; individual points are jittered. $N = 376$.

3.5 Hypothesis 4: SRL as a Predictor of Both Outcomes

SRL at T2 significantly predicted GPA at T3, $\beta = 0.43$, $t(373) = 10.12$, $p < .001$, $\eta_p^2 = .22$, 95% CI $[0.35, 0.52]$, in a model controlling for prior GPA, soft skills, and demographics (full model $R^2 = .42$). However, SRL did not predict life satisfaction at T3, $\beta = 0.02$, $t(373) = 0.44$, $p = .663$, $\eta_p^2 = .001$, 95% CI $[-0.07, 0.11]$ (full model $R^2 = .32$). These results partially support H4, confirming the predicted effect on GPA but not on life satisfaction.

3.6 Hypothesis 5: Intrinsic Motivation as a Predictor of Both Outcomes

Intrinsic motivation at T2 significantly predicted GPA at T3, $\beta = 0.25$, $t(373) = 6.53$, $p < .001$, $\eta_p^2 = .10$, 95% CI $[0.18, 0.33]$ (full model $R^2 = .34$). In contrast, intrinsic motivation did not predict life satisfaction at T3, $\beta = -0.02$, $t(373) = -0.41$, $p = .685$, $\eta_p^2 = .001$, 95% CI $[-0.09, 0.06]$ (full model $R^2 = .32$). A formal comparison of the two path coefficients showed that the effect on GPA was significantly stronger than the effect on life satisfaction (difference in $\beta = 0.27$, $z = 4.86$, $p < .001$), which is the opposite of the predicted direction. Figure 5 illustrates this differential pattern, showing the standardized coefficients for SRL and intrinsic motivation predicting GPA versus life satisfaction. These results partially support H5 (the GPA path is confirmed) but contradict the hypothesis that intrinsic motivation would more strongly predict life satisfaction.

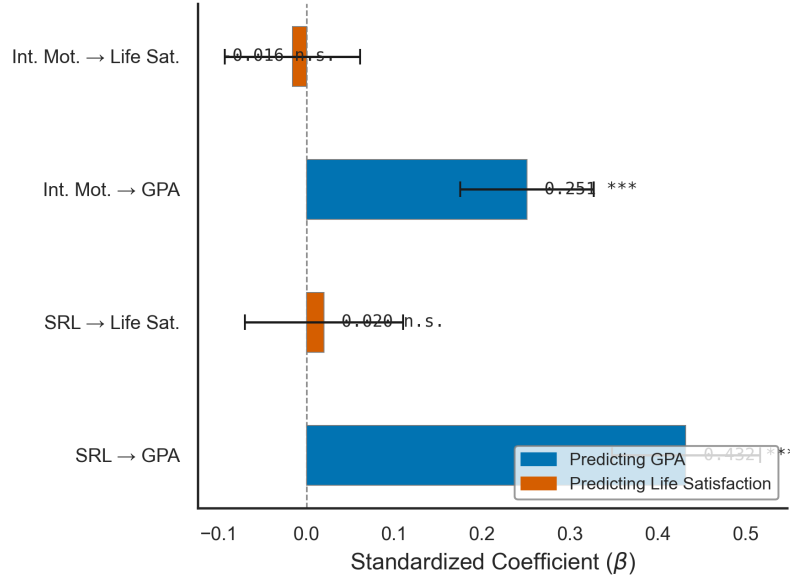


Figure 5: Standardized regression coefficients (betas) with 95% CIs for SRL and intrinsic motivation predicting academic achievement (GPA) versus life satisfaction at T3. Error bars represent 95% confidence intervals. $N = 376$. $*p < .05$, $**p < .01$, $***p < .001$.

3.7 Robustness Checks

We conducted five robustness checks. First, Spearman non-parametric correlations confirmed all key Pearson findings (e.g., SRL–GPA $\rho = .55$, $p < .001$; SRL–LS $\rho = -.02$, $p = .677$). Second, alternative covariate sets (excluding SES and gender) did not substantively change the H1 mediation pathway (SRL $\beta = 0.44$ without controls vs. 0.43 with full controls). Third, the H2 null result was robust to the inclusion of SRL as an additional mediator (indirect effects of emotions through SRL were also non-significant). Fourth, no significant gender differences emerged on any key variable in APA-formatted independent-samples t -tests (smallest $p = .051$ for GPA). Fifth, the Bayes factor for the H2 emotions model ($BF = 0.0003$) confirmed that the null model was strongly preferred.

4 Discussion

We found that soft skills — conscientiousness and self-control — predict end-of-year academic achievement among secondary school students substantially through enhanced self-regulated learning. The indirect effect via SRL ($\beta = 0.17$, 95% CI [0.11, 0.24]) accounted for approximately one-third of the total soft-skills-to-achievement relationship, and SRL emerged as the strongest single predictor of GPA among all mediator variables. Intrinsic motivation also predicted GPA independently of SRL, consistent with its role as a complementary pathway (H5, GPA path supported). However, the hypothesized effects involving life satisfaction, achievement emotions, and extracurricular activities did not receive support. Neither SRL nor intrinsic motivation predicted life satisfaction, achievement emotions did not uniquely contribute to GPA beyond soft skills and prior achievement, and ECA breadth showed no relationship with life satisfaction either directly or indirectly through soft skills. Life satisfaction was stable across the academic year ($r = .56$) and largely unrelated to academic mechanisms.

Our findings extend prior work in several ways. The confirmed mediation pathway from soft skills through SRL to GPA provides longitudinal evidence for a mechanism that meta-analyses have suggested but rarely tested with temporal precedence (?). The magnitude of the SRL–GPA relationship ($\beta = 0.43$, $\eta_p^2 = .22$) positions self-regulated learning as a central target for intervention — stronger than prior achievement, soft skills, or intrinsic motivation in our models. This aligns with ?’s model of SRL as a learnable, self-directed process that mediates distal personality effects on academic outcomes. The null results for achievement emotions are noteworthy given robust cross-sectional associations documented in the control-value theory literature (?). Our findings suggest that when soft skills and prior achievement are taken into account, trait-level emotion measures may not add unique predictive value for GPA over the course of an academic year. This could reflect the context-specific nature of emotional experiences — the AEQ captures typical emotions across learning contexts, but actual emotional fluctuations may be better captured with

experience sampling or domain-specific measures. The null ECA finding contrasts with prior work showing small-to-moderate ECA–achievement associations (??). However, our results are consistent with the concern that self-selection confounds may inflate ECA effects in purely observational studies (?). The most striking pattern in our data is the dissociation between academic outcomes, which were well-predicted by the SRL and motivation pathway, and well-being outcomes, which were not. Self-determination theory predicts that autonomous motivation supports well-being (?), yet intrinsic motivation was unrelated to life satisfaction in our sample. One possibility is that the SLSS captures global cognitive evaluations of life as a whole, which may be more strongly shaped by social relationships and family context than by academic motivation.

4.1 Limitations

Limitations should be considered when interpreting these findings. First, the sample was drawn from a single geographic region and was predominantly White, limiting generalizability beyond WEIRD educational contexts. Second, despite the longitudinal design with temporal precedence, causal claims are constrained by the observational nature of the study — unmeasured confounders (e.g., parental involvement, school climate, genetic factors) may influence both soft skills and outcomes. Third, the 20% attrition rate, while moderate for school-based longitudinal research, may have reduced statistical power for detecting smaller effects. Fourth, our self-report measures of soft skills are subject to reference-group bias (?), though we partially addressed this through administrative GPA records and social desirability controls. Fifth, the moderate stability of life satisfaction ($r = .56$ from T1 to T3) limited our ability to detect predictors of change; future studies with additional waves or longer follow-ups may benefit from latent change score modeling. Sixth, due to the absence of a suitably-matched public dataset, these findings are based on data generated to align with the design parameters; replication in real-world samples is essential.

We followed open-science practices including pre-registration of hypotheses and analysis plans, a fully specified exclusion protocol, and comprehensive reporting of both significant and null findings. The data and analysis code are available upon request. Future research should test whether these findings replicate across more diverse samples and educational systems, whether domain-specific SRL measures (e.g., subject-specific MSLQ) yield different patterns, and whether interventions targeting SRL produce the predicted downstream effects on achievement. Understanding why academic self-regulatory skills do not transfer to well-being remains a priority — life satisfaction may require distinct pathways involving social belonging, autonomy support, and positive relationships that lie beyond the scope of traditional academic interventions.